

1. Business Description -

Currently, the bike rental company operates with a traditional demand forecasting approach. Rentals are influenced by season, weather, holidays, and working days, but predictions are made manually or using basic historical averages. This often leads to mismatches between supply and demand, especially during peak seasons or unexpected weather changes.

2. Problems Faced –

- Difficulty in accurately predicting demand, leading to either shortage of bikes or excess idle inventory.
- Seasonal fluctuations cause inconsistent revenue streams.
- Weather conditions (rain, humidity, wind) reduce rentals unpredictably.
- Lack of data-driven insights for marketing and resource allocation.

3. Possible Use Case for AI

AI can be implemented to build predictive models using historical rental data, weather forecasts, and calendar variables.

- Machine learning algorithms (like Linear Regression, Ridge/Lasso) can forecast daily demand with high accuracy.
- AI can recommend dynamic pricing strategies based on demand.
- AI can optimize bike distribution across locations to reduce shortages.
- AI can provide insights for marketing campaigns (e.g., promotions during low-demand periods).

4. Business Description – Post AI Implementation

After AI implementation, the company will operate with a data-driven approach. Demand forecasting will be automated, reducing errors and improving customer satisfaction. Inventory will be optimized, ensuring bikes are available when needed. Marketing will be more targeted, increasing revenue during off-peak times. Overall, AI will transform the business into a smarter, more efficient, and customer-centric operation.